

MASTER OF SCIENCE IN PRECISION SCIENTIFIC INSTRUMENTATION (MSPSI) DISSERTATION

Program Director:	Assoc Prof Wang Xiao, Renshaw
Email add:	renshaw@ntu.edu.sg

AGENDA

- OBJECTIVES
- DURATION & CONTACT HOURS
- SIGN UP TIMELINE
- EXAMINATION AND DELIVERABLES
- FAQ AND Q&A

DISSERTATION OBJECTIVES

- Training on necessary concepts and skills related to advanced scientific instrumentation
- Weekly Consultations with supervisors
- Plan and execute a research project
- Carry out analysis
- Write and submit Dissertation for Examination
- Take Oral Examination to defend Dissertation
- Dissertation will not have letter grade, and the title will be reflected on Student's Transcript

DURATION & CONTACT HOURS

- Start Early February 2026
- **AT LEAST SIX months duration**, extension/postponement is subjected to approval by Supervisor and Programme Director
- Weekly consultations with Project Supervisor

EXAMINATION SCHEDULE

Examination	Length	Submission Deadline
Dissertation (1 st version)	60 - 100 pages	Full time: <u>No earlier</u> than 3 rd August 2026 Part time: <u>No earlier</u> than 1 st February 2027
Release of Dissertation Examination Outcomes (1 st version)	-	1 month* after submission
Dissertation (Final version)	60 - 100 pages	Full time: <u>No later</u> than 4 months before maximum candidature of 2 years Part time: <u>No later</u> than 4 months before 3 years due
Release of Dissertation Examination Outcomes (Final version)	-	2 Weeks* after submission
Oral Examination		To be arranged with Examiners

* Students to communicate with Supervisor and Examiners with regards to examination timelines

DISSERTATION

- 60 to 100 pages
- Must include:
 1. Title Page,
 2. Statement of Originality,
 3. Supervisor Declaration Statement,
 4. Authorship Attribution Statement,
 5. References
- Include iThenticate Report with **LESS THAN 20% Similarity** and **LESS THAN 30% Artificial Intelligence Index**
- MSPSI Dissertation Guideline available here:
[SPMS Guideline MSc Dissertation.pdf](#)

ORAL EXAMINATION

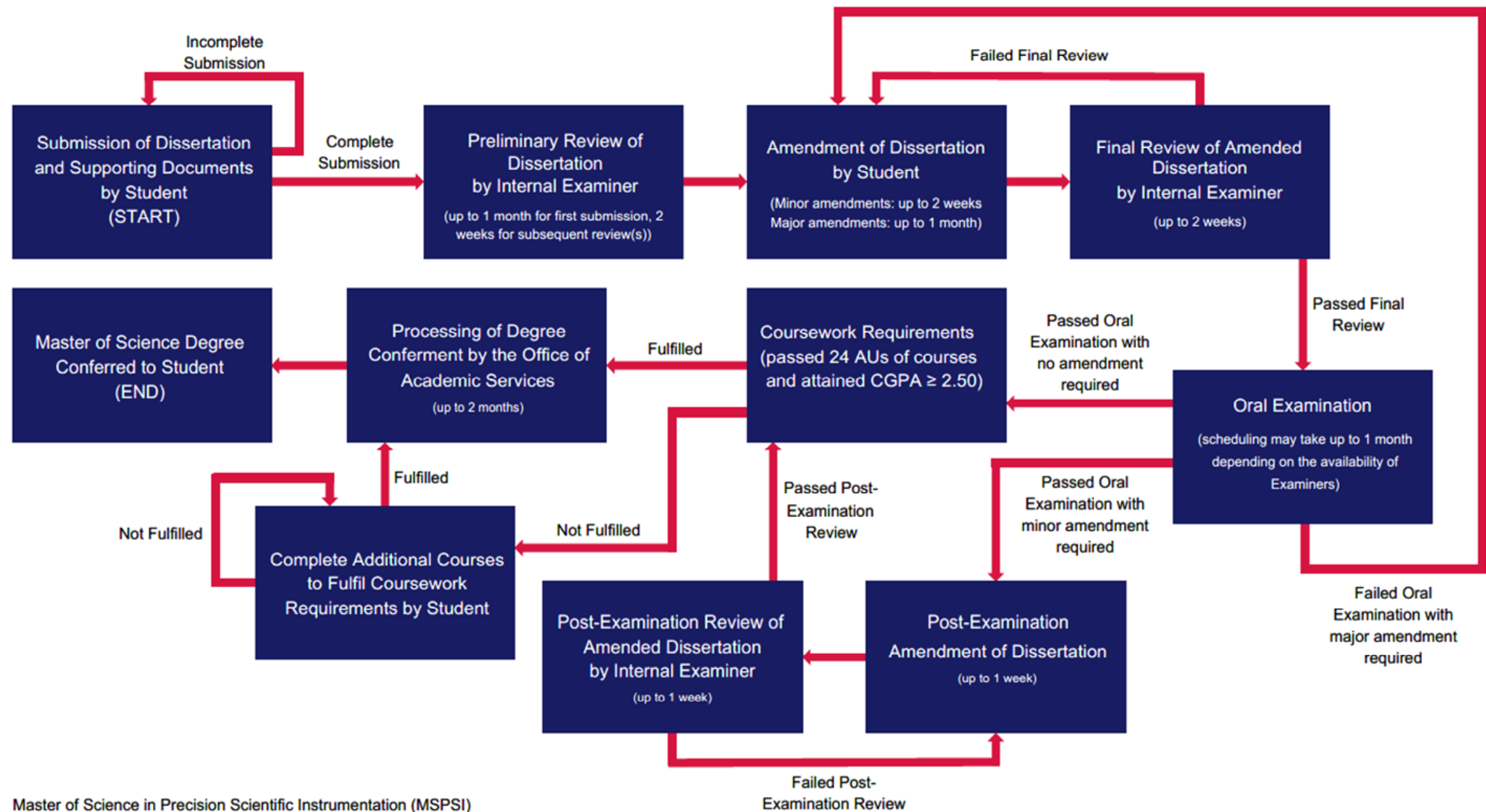
- Submit finalised copy of Presentation Slides to MSPSI and Examiners **AT LEAST ONE DAY** before Examination
- Must include:
 1. Dissertation Project Title
 2. Specific aim(s) of Dissertation Project
 3. Significance of Dissertation Project
 4. Approach of Dissertation Project
 5. Innovation Outcome from Dissertation Project
 6. List of References

SIGN UP SCHEDULE

Description	Date	
Dissertation topics Emailed to Students	15 th January 2026	
Students view available project topics and approach supervisors (students can arrange with supervisors directly*) for topic discussions	15 th January 2026 – 21 st January 2026	
Supervisor allocates project to student	16 th January 2026 – 23 rd January 2026	
Students change Option of Study with OAS	The list of students change request will be submitted for approval and processing <u>AFTER</u> the project allocation period has ended.	
Commencement of Dissertation Project	24 th January 2026 onwards	
MINIMUM allowable period (EARLIEST) to submit fully completed dissertation for examination	Full-time	3 rd August 2026
	Part-time	1 st February 2027
MAXIMUM allowable period (LATEST) to submit fully completed dissertation for examination	Full-time	4 months before end of 2nd Year Candidature
	Part-time	4 months before end of 2nd Year Candidature

* If interested, students are welcomed to reach out to prospective Supervisors within **Physics Department** to discuss and produce additional topics. Each Supervisor can **ONLY SUPERVISE UP TO 3 students**.

Dissertation Examination Process Flow



ESTIMATED TIMELINE

Submission for Examination	Release of Dissertation Examination Outcome (estimated)	Submission of the Final Version (only applicable for No Revision or Minor Revision)	Graduation Date to be printed on the Degree Certificate	Collection of Degree Certificate and Official Transcript from One Stop @ SAC
Apr	Jun	Before End-Jun	End-Jul	Mid-Aug
May	Jul	Before End-Jul	End-Aug	Mid-Sep
Jun	Aug	Before End-Aug	End-Sep	Mid-Oct
Jul	Sep	Before End-Sep	End-Oct	Mid-Nov
Aug	Oct	Before End-Oct	End-Nov	Mid-Dec
Sep	Nov	Before End-Nov	End-Dec	Mid-Jan
Oct	Dec	Before End-Dec	End-Jan	Mid-Feb
Nov	Jan	Before End-Jan	End-Feb	Mid-Mar
Dec	Feb	Before End-Feb	End-Mar	Mid-Apr
Jan	Mar	Before End-Mar	End-Apr	Mid-May
Feb	Apr	Before End-Apr	End-May	Mid-Jun
Mar	May	3 rd week-May	End-Jun	Mid-Jul

DISSERTATION TOPICS – 1/3

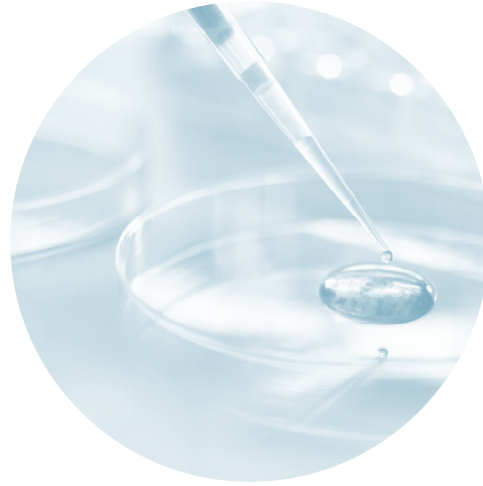
No.	Supervisor	Research Projects	Faculty Email Address
1	Assoc Prof S.N. Piramanayagam	He ion irradiation induced magnetic property modulations	prem@ntu.edu.sg
2	Assoc Prof S.N. Piramanayagam	Enhancement of spin-orbital torque efficiency by tuning the crystal structures of the orbital Hall layer	prem@ntu.edu.sg
3	Assoc Prof S.N. Piramanayagam	Exploring Spin-Orbit-Torque (SOT) and Dzyaloshinskii-Moriya-interaction (iDMI) in Thin-films for Spintronic Applications	prem@ntu.edu.sg
4	Assoc Prof S.N. Piramanayagam	3D magnetic field sensor driven by spin-orbit torque	prem@ntu.edu.sg
5	Asst Prof Shen Yijie	Topological optical skyrmion generation by digital holograms	yijie.shen@ntu.edu.sg
6	Asst Prof Shen Yijie	Generation of Optical Superoscillatory Modes Using Digital Micromirror Devices (DMDs)	yijie.shen@ntu.edu.sg
7	Asst Prof Shen Yijie	Simulation and Control of Optical Toroidal Resonances and Anapole Modes in On-Chip Systems	yijie.shen@ntu.edu.sg
8	Asst Prof Shen Yijie	Axion electrodynamics and axion field calculation	yijie.shen@ntu.edu.sg
9	Asst Prof Shen Yijie	Propagation-Invariant Topological Vortices in Water Waves via Metamaterials	yijie.shen@ntu.edu.sg
10	Asst Prof Shen Yijie	Acoustic Vortex Tweezers at Quasi-Bound States in the Continuum (QBIC)	yijie.shen@ntu.edu.sg
11	Prof Chew Lock Yue	Nonequilibrium thermodynamics of quantum information ratchets driven by quantum algorithms	lockyue@ntu.edu.sg
12	Prof Fan Hongjin	Advanced High Entropy Electrolyte for aqueous Zinc-ion battery	FANHJ@ntu.edu.sg
13	Prof Fan Hongjin	Printed Paper Batteries	FANHJ@ntu.edu.sg
14	Prof Fan Hongjin	Fabrication and test of rechargeable Zn-Cu batteries	FANHJ@ntu.edu.sg
15	Dr Gerard Joseph Lim	Integrating LiDAR and Machine Learning on Low-Cost ARM Microcontrollers for Autonomous Navigation	gerardjosephlim@ntu.edu.sg
16	Prof Gao Weibo	Spin-Active Near-Infrared Quantum Emitters in Hexagonal Boron Nitride	wbgao@ntu.edu.sg
17	Prof Gao Weibo	Tuning the moiré superlattices at extreme conditions	wbgao@ntu.edu.sg
18	Prof Gao Weibo	Topological Domain Structure in Van Der Waals Ferroelectric	wbgao@ntu.edu.sg
19	Prof Gao Weibo	Multi-fidelity modeling for 2D quantum excitonic devices	wbgao@ntu.edu.sg
20	Prof Gao Weibo	Separating depolarization field and BPVE photocurrent in 3R-MoS2 ferroelectric devices	wbgao@ntu.edu.sg

DISSERTATION TOPICS – 2/3

No.	Supervisor	Research Projects	Faculty Email Address
21	Prof Gao Weibo	Spin-orbit coupling in 2D materials for optical vortex generation	wbgao@ntu.edu.sg
22	Prof Gao Weibo	Degree of Valley Polarization in TMD interacted with Twisted Light	wbgao@ntu.edu.sg
23	Prof Gao Weibo	Tuning spin-dependent phenomena via strain in g-wave altermagnets	wbgao@ntu.edu.sg
24	Prof Gao Weibo	Enhancing Quantum Sensing Sensitivity via 1042 nm Infrared Readout of Diamond NV Centers	wbgao@ntu.edu.sg
25	Prof Gao Weibo	Investigation of Orbital Angular Momentum Generation in van der Waals Materials	wbgao@ntu.edu.sg
26	Prof Gao Weibo	Rhombohedral Graphene for Exploring Correlated Topological States	wbgao@ntu.edu.sg
27	Prof Gao Weibo	Electrically Tunable Quantum Anomalous Hall States in Twisted MoTe2	wbgao@ntu.edu.sg
28	Prof Gao Weibo	Flat Band Physics in Rhombohedral Stacked Graphene	wbgao@ntu.edu.sg
29	Prof Gao Weibo	Polaritonic bistability in Monolayer WS2 at room temperature	wbgao@ntu.edu.sg
30	Prof Gao Weibo	Large-scale high-quality exfoliation of two-dimensional monolayers	wbgao@ntu.edu.sg
31	Prof Gao Weibo	Exciton Lifetime as a Probe of Interlayer Correlation in Moiré Excitonic Insulators	wbgao@ntu.edu.sg
32	Prof Gao Weibo	Structure-property correlations in quantum anomalous Hall devices	wbgao@ntu.edu.sg
33	Prof Gao Weibo	Anisotropic van der Waals Metasurfaces for Tunable Entangled Photon-Pair Generation	wbgao@ntu.edu.sg
34	Assoc Prof Yang Bo	Modern machine learning approach for classical many-body systems with applications for logistics optimizations	yang.bo@ntu.edu.sg
35	Assoc Prof Yang Bo	Dynamics of Anyons in 2D strongly correlated topological materials	yang.bo@ntu.edu.sg
36	Assoc Prof Yang Bo	Reinforcement learning for logistics engineering and optimizations	yang.bo@ntu.edu.sg
37	Assoc Prof Yang Bo	Reinforcement learning for elevator and shuttle bus service optimizations	yang.bo@ntu.edu.sg
38	Prof Wang Qijie	Terahertz quantum cascade laser characterizations	qjwang@ntu.edu.sg
39	Prof Wang Qijie	Multiple compound gas sensing with mid-infrared laser technologies	qjwang@ntu.edu.sg
40	Prof Wang Qijie	Food inspection using high performance quantum cascade lasers	qjwang@ntu.edu.sg

DISSERTATION TOPICS – 3/3

No.	Supervisor	Research Projects	Faculty Email Address
41	Prof Wang Qijie	Graphene and 2D material-based photodetector and optoelectronic devices	qjwang@ntu.edu.sg
42	Prof Wang Qijie	2D material-based photodetector for mid-infrared detection	qjwang@ntu.edu.sg
43	Prof Wang Qijie	2D material-based optoelectronic devices	qjwang@ntu.edu.sg
44	Prof Wang Qijie	Mid-infrared quantum cascade laser characterizations	qjwang@ntu.edu.sg
45	Prof Wang Qijie	Edible oil identification using high-performance quantum cascade lasers and FTIR spectroscopy	qjwang@ntu.edu.sg
46	Prof Wang Qijie	Nonlinear optics in 2D materials	qjwang@ntu.edu.sg
47	Prof Wang Qijie	Mid-infrared silicon photonics	qjwang@ntu.edu.sg
48	Prof Wang Qijie	2D nanomaterials for nonlinear optics	qjwang@ntu.edu.sg
49	Assoc Prof David Wilkowski	Precision Measurement on an Atomic Wavepacket Interferometry	david.wilkowski@ntu.edu.sg
50	Assoc Prof David Wilkowski	Quantum simulator based on ultracold atomic gas	david.wilkowski@ntu.edu.sg
51	Assoc Prof WANG Xiao, Renshaw	Neuromorphic Electronics Based on Ferroelectric Field-Effect Transistors	renshaw@ntu.edu.sg
52	Assoc Prof WANG Xiao, Renshaw	Optoelectronic Properties of Ionic Field-Effect Transistors	renshaw@ntu.edu.sg
53	Assoc Prof WANG Xiao, Renshaw	Spectroscopic Measurement of Low-Dimensional Systems	renshaw@ntu.edu.sg
54	Assoc Prof WANG Xiao, Renshaw	Mechanical Resonance of Atomically Thin Systems	renshaw@ntu.edu.sg



Q&A



MSPSI

FAQ

- o Can I have a template?
Yes, template has been provided.
- o Can I do the research project remotely; in my home country? No, you need to remain in SG during the duration of studies unless Leave of Absence is obtained.
- o For the oral presentation, will I present to my supervisor?
No, you will present to two independent examiners that are assigned to you.
- o Will graduation date be stated on the Master's certificate?
Yes, your Master's certificate will have graduation date stated. The printed graduation date will depend on when the Final Version of the Dissertation has been submitted.
- o When will you be receiving the graduation certificate?
You may visit the link to find out more:
[Graduation Matters | NTU Singapore | NTU Singapore](#)